

open source in its design as well as affordable to make, procure and use. This would let them deal with projects of higher complexity and finer levels of control. What they were looking for, was a micro-controller.

A microcontroller is like a mini-CPU on a chip that contains a processing core, memory and programmable input/output ports. They were originally designed to be used as embedded computers, or computers within other devices, electronic and non-electronic.

The maker community was waiting for a micro-controller that fell within the following criterion:

1. Something that could easily interface with the electronics and hardware in most common devices easily.
2. Even though it would do tasks usually done on full scale computers, it will do them at a fraction of the cost.
3. It could be reused and modified as according to usage.

Since 1971, Intel has been bringing microcontrollers to the markets, which were primarily used with external chips to implement a functional computer system. Most of the early microcontrollers didn't even come close to satisfying any of the criteria that could make them suitable for DIY or Maker Tech.

Fast forward to 2006. Components for integrated chips are coming at the cheapest rates ever. And thanks to Open Source Software, there are purpose-specific programming languages for almost any possible usage. For example, take Processing.

Initiated back in 2001 by Casey Reas and Benjamin Fry formerly of the MIT Media Lab, it is an open source programming language and IDE mainly built with the target of getting non-programmers to program, it works mostly in the visual context and is inspired from Logo along with Java and OpenGL and provides visual feedback to programming. Building on this language, the Wiring platform was developed in 2003. It consisted of a programming language, an IDE along with a tied-in single-board microcontroller. Although this did not create as much popularity in the Maker community as the boards that came later, one can say that the Wiring platform was the silence before the storm that was about to enter the Maker-DIY community.

Birth of the Game Changer – Arduino microcontroller

The co-founder of the Arduino Project, Massimo Banzi had one goal – how to get his students at the Interaction Design Institute, Ivrea (IDII) to make